

Development of neuro-fuzzy model to explore gene–nutrient interactions modulating warfarin dose requirement

Aim: To investigate the influence of alterations in vitamin K (K1, K2 and K3) in modulating warfarin dose requirement. **Patients & methods:** Reverse phase HPLC to determine the plasma vitamin K; PCR-RFLP to detect polymorphisms; and the neuro-fuzzy model to predict warfarin dose were used. **Results:** The developed neuro-fuzzy model showed a mean absolute error of 0.000024 mg/week. *CYP2C9**2 and *CYP2C9**3 mediated warfarin sensitivity was observed when vitamin K is in high and low tertiles, respectively. *VKORC1*-1639G>A exhibited warfarin sensitivity in all combinations. Higher vitamin K1 was observed in *CYP4F2* V433M polymorphism. The requirement of warfarin is low in GGCX 8016 GG genotype compared with GA and AA genotypes. **Conclusion:** Vitamin K profile along with genetic testing ensures precision in warfarin dose optimization.

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Keywords: *CYP2C9* • INR • neuro-fuzzy model • vitamin K • *VKORC1* • warfarin

Background

Warfarin inhibits VKORC1 therefore preventing the activation of vitamin K-dependent clotting factors of extrinsic coagulation pathway, in other words, factors II, VII, IX and X; and proteins C, S and Z [1]. Hence, it is widely prescribed for the prevention of thrombotic events. The liver metabolizes *R*- and *S*-enantiomers of warfarin using CYP1A1, CYP1A2, CYP3A4 and CYP2C9 enzymes to form 6-, 8-, 10- and 7-hydroxy warfarin, respectively [2]. Since *S*-warfarin is the most potent inhibitor of VKORC1, CYP2C9 activity dictates the efficacy of warfarin dose required for effective coagulation. In poor metabolizers of CYP2C9, in other words, *CYP2C9**2 (R144C) and *CYP2C9**3 (I359L), the warfarin to 7-OH warfarin ratio was found to be higher compared with wild genotype and hence requiring a lower dose of warfarin [3,4]. Subjects with *VKORC1*-1639G>A exhibit lower warfarin requirement due to lower expression of VKORC1 [5].

Several pharmacogenomic algorithms were developed taking into account demographic and genetic variables to optimize the warfarin dose and to minimize adverse drug reactions [6–8]. These algorithms did not include the influence of changes in plasma vitamin K on the warfarin dose requirements. Vitamin K1 known as phyloquinone, is the predominant form of vitamin K in the diet coming from plants. Vitamin K2 known as menaquinone, is synthesized by the human intestinal microbiota and also presents in certain animal products. Intestinal microbiota can also convert vitamin K1 to K2. Vitamin K3 known as menadione, is a synthetic provitamin, which needs to be converted to menaquinone to be active. Vitamin K hydroquinone (KH2) acts as a cofactor for GGCX to catalyze the carboxylation of specific glutamic acid residues in vitamin K-dependent clotting factors to facilitate calcium binding. To enable this reaction, KH2 is oxidized to form vitamin K

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epoxide (KO). The VKORC1 catalyzes the reduction of KO to vitamin K thus contributing to recycling of vitamin K. Warfarin inhibits VKORC1 resulting in functional vitamin K deficiency. Increased intake of dietary or supplemental vitamin K can negate the anticoagulant effect of warfarin. The current study aimed to elucidate such influences on warfarin therapy using a neuro-fuzzy model.

Neuro-fuzzy model is a hybrid artificial intelligence system with a combination of human-like reasoning of fuzzy systems for learning and connectionist architecture of neural networks. This model is capable of meeting two contradictory requirements, in other words, interpretability and accuracy. The neuro-fuzzy systems are considered as universal approximators with the ability to interpret based on IF THEN rules.

Materials & methods
Study design & participants

A total of 115 patients (63 men and 52 women) in the age group of 12–75 years who were on warfarin therapy for prevention of thrombotic events were enrolled for the study. All these patients underwent valve replacement therapy. The study protocol was approved by the Institutional Ethical Committee of Nizam’s Institute of Medical Sciences, Hyderabad, India (EC/NIMS/1170/2010). An informed consent was obtained from all the patients. The demographic details, namely age, gender, BMI, diet, among others, were recorded through the personal interviews. The international normalized ratio (INR) values and history of adverse drug reactions were recorded based on the medical records (Table 1). Patients who require warfarin <28 mg/week were considered as ‘warfarin sensitive,’ those with requirement between 28 to 49 mg/week were labeled as ‘warfarin intermediate’ and when the requirement was >49 mg/week patients were labeled as ‘warfarin resistant.’ All the patients included

in the current study had INR ranging from 2 to 3.5 for three consecutive follow-ups spanning 3 months period. Out of the enrolled patients, only one patient was being treated with thyronorm, which is likely to influence warfarin therapy. None of the other patients were on any other medication likely to influence warfarin therapy. Whole blood samples were collected in EDTA after an overnight fast to exclude the influence of recent dietary intake.

Genetic analysis
PCR-RFLP approach was used to detect *CYP2C9**2, *CYP2C9**3, *CYP2C9**8, *VKORC1*-1639G>A, *VKORC1**3, *VKORC1**4, *CYP4F2* V433M, *GGCX* G8016A polymorphisms as described earlier [9].

Determination of vitamin K (K1, K2, K3) by HPLC
Reverse phase HPLC was used for the resolution and determination of vitamin K1, K2 and K3 as described earlier [9]. The estimated values of these vitamins were distributed as three tertiles as shown in Table 1.

Development of neuro-fuzzy logic model
For the development of neuro-fuzzy logic model, we have used age, gender, BMI, *CYP2C9**2, *CYP2C9**3, *VKORC1*-1639 G>A, *CYP4F2* V433M, *GGCX* G8016A, thyroid status, vitamin K1, K2 and K3 levels of plasma as the input variables. In order to increase the performance of the model, vitamin K levels were converted into three tertiles (Table 2). The therapeutic dose of warfarin when the INR was between 2.0 and 3.5 was used as the output variable. Fuzzy Interference System was generated using the sub clustering. Subclustering was performed by optimizing the range of influence as 0.5, squash factor as 1.25, accept ratio as 0.5 and reject ratio as 0.15. The model was trained based on the data of 115 patients who received an optimal dose of warfarin. The Fuzzy Interference

Table 1. Patient characteristics.	
Parameter	Mean ± SD
Age (years)	39.03 ± 14.51
Male:female ratio	63:52
Body mass index (Kgm ⁻²)	21.66 ± 4.27
<i>CYP2C9</i> *2 (wild:hetero)	91:24
<i>CYP2C9</i> *3 (wild:hetero)	94:21
<i>VKORC1</i> -1639 (wild:hetero:homo)	82:32:1
<i>CYP4F2</i> V433M (wild:hetero:homo)	30:61:24
<i>GGCX</i> G8016A (wild:hetero:homo)	87:27:1
SD: Standard deviation.	

Table 2. The distribution of vitamin K in the plasma of patients being treated with warfarin.			
Vitamin K isoform	Tertile 1	Tertile 2	Tertile 3
Vitamin K1 (ng/ml)	0.12–28.05	>28.45–163.72	>163.72–1008.85
Vitamin K2 (ng/ml)	0.41–52.10	>52.10–139.96	>139.96–9065.02
Vitamin K3 (ng/ml)	4.91–236.36	>236.36–521.01	>521.01–31489.37

System optimization for the training of the model was based on ‘hybrid’ method with error tolerance of 0.0001 and epochs of 3000. The training of the model was stopped when the mean absolute error was minimum. The performance of the model was ascertained by cross validating the data as the testing and the checking data. A total of 115 rules were generated to build this model. The final model depicts rules and surface plots representative of interactions between the variables.

Results

The neuro-fuzzy model showed a mean absolute error of 0.000029. As shown in **Figures 1 & 2**, the association of *CYP2C*3* and *VKORC1*-1639G>A with warfarin sensitivity was evident irrespective of the vitamin K status. The positive association of warfarin dose with vitamin K was evident, specifically with vitamin K3 in subjects with *VKORC1*-1639G>A polymorphism. Similar positive association was observed even with vitamin K1 and K2 except for the first tertile in these subjects. Subjects with *CYP2C9*3* polymorphism exhibited positive association between vitamin K1 and warfarin requirement. However, increase in vitamin K2 and K3 does not affect the warfarin sensitivity of *CYP2C9*3* variant.

As shown in **Figure 3**, *CYP2C9*2* variant exhibited warfarin sensitivity when vitamin K1 and K3 levels were in the second and third tertiles; and vitamin K2 levels were in the first tertile. Positive association between warfarin and vitamin K was observed only with K2 and K3.

When the vitamin K1 levels were in the lowest tertile, the requirement of warfarin was found to be much lower in *CYP4F2* CC and CT genotypes. The *CYP4F2* TT genotype showed a positive association between vitamin K1 and warfarin dose. The presence of C-allele decreased this requirement. A similar trend was observed with changes in vitamin K2 and K3 levels (**Figure 4**).

The requirement of warfarin was lower in subjects with *GGCX* G allele compared with those with A allele. The warfarin requirements were higher in subjects whose vitamin K3 levels were in the highest tertile (50 vs 20 mg/week for GG). A similar trend was observed with vitamin K1 and K2 (30 vs 10 mg/week for GG) (**Figure 5**).

Discussion

The pharmacogenomic algorithms available for warfarin dose optimization were based on multiple linear regression [8,10–13], polynomial regression [14] and artificial neural network [9,15,16]. These algorithms showed a mixed response in clinical trials [17,18]. One possible contributor toward this mixed response could be altered vitamin K profile, which significantly modulates warfarin dose requirements. The novelty in the current study was the application of neuro-fuzzy logic model to delineate the gene–nutrient interactions with respect to vitamin K1, K2 and K3. This model showed excellent efficacy as evidenced by the narrow mean absolute error. Furthermore, it highlighted how alterations in vitamin K isoforms influence intraindividual variations in warfarin dose requirement.

The intake of vitamin K in warfarin patients was found to be acceptable in the range of 25–325 µg/day [19]. Plasma vitamin K levels were reported to decrease after the initiation of warfarin treatment [20]. More therapeutic monitoring was needed for the patients with low plasma vitamin K [20]. Better INR control was observed in patients with higher vitamin K intake [21]. Higher plasma vitamin K1 and menaquinone-4 were observed in subjects with *CYP4F2* rs2108622 TT genotype compared with dose with CC and CT genotype [22]. Higher plasma phylloquinone levels were reported in subjects with *VKORC1* rs8050894 GG genotype compared with those with CG or CC genotypes [23]. Subjects with *VKORC1* rs7294 AA or AG genotype showed lower plasma phylloquinone compared to those with GG genotype [23].

Our results are consistent with existing literature in demonstrating the association of *CYP2C9*2* and *CYP2C9*3* polymorphisms with warfarin sensitivity. However, the current study adds a new dimension for this association by demonstrating warfarin sensitivity of *CYP2C9*2* only when vitamin K1 levels are in second and third tertiles (high) and K2 and K3 were low. *CYP2C9*3* induced warfarin sensitivity was observed when vitamin K1 levels are in the first and second tertiles or K2 and K3 are high.

The *VKORC1*-1639G>A polymorphism was reported to abolish the E-box consensus sequences in the promoter and hence compared with A-allele, G allele was

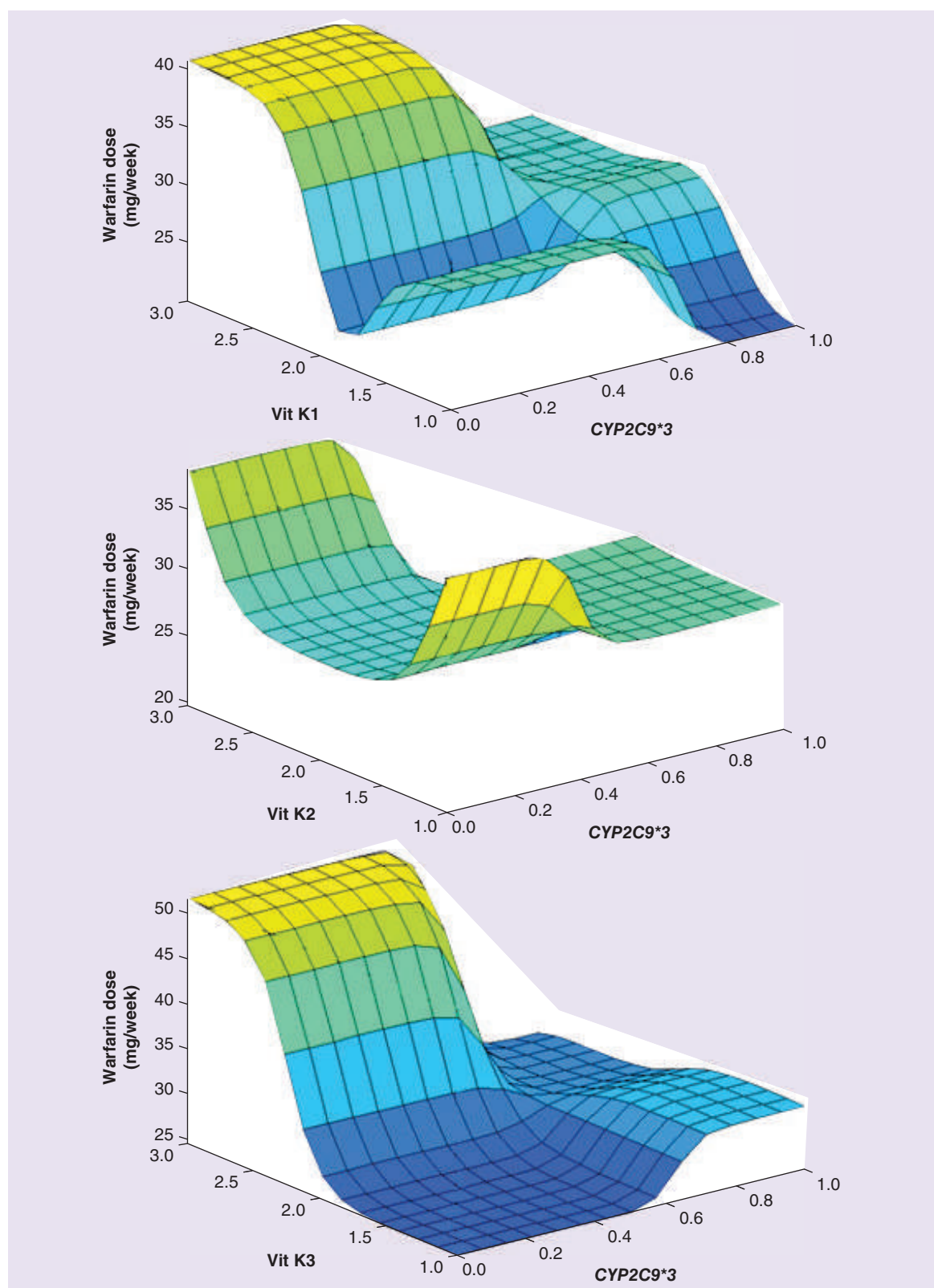


Figure 1. Association of warfarin dose with CYP2C9*3 and vitamin K1, K2, K3 levels. (A) CYP2C9*3 mediated warfarin sensitivity is more pronounced only in subjects with vitamin K1 in the lowest tertile only in mutants. (B) CYP2C9*3 mediated warfarin sensitivity is found to more pronounced with vitamin K2 in highest tertile. (C) In CYP2C9*3 wild-type, warfarin requirement is more in third tertile in which vitamin K3 levels are higher.

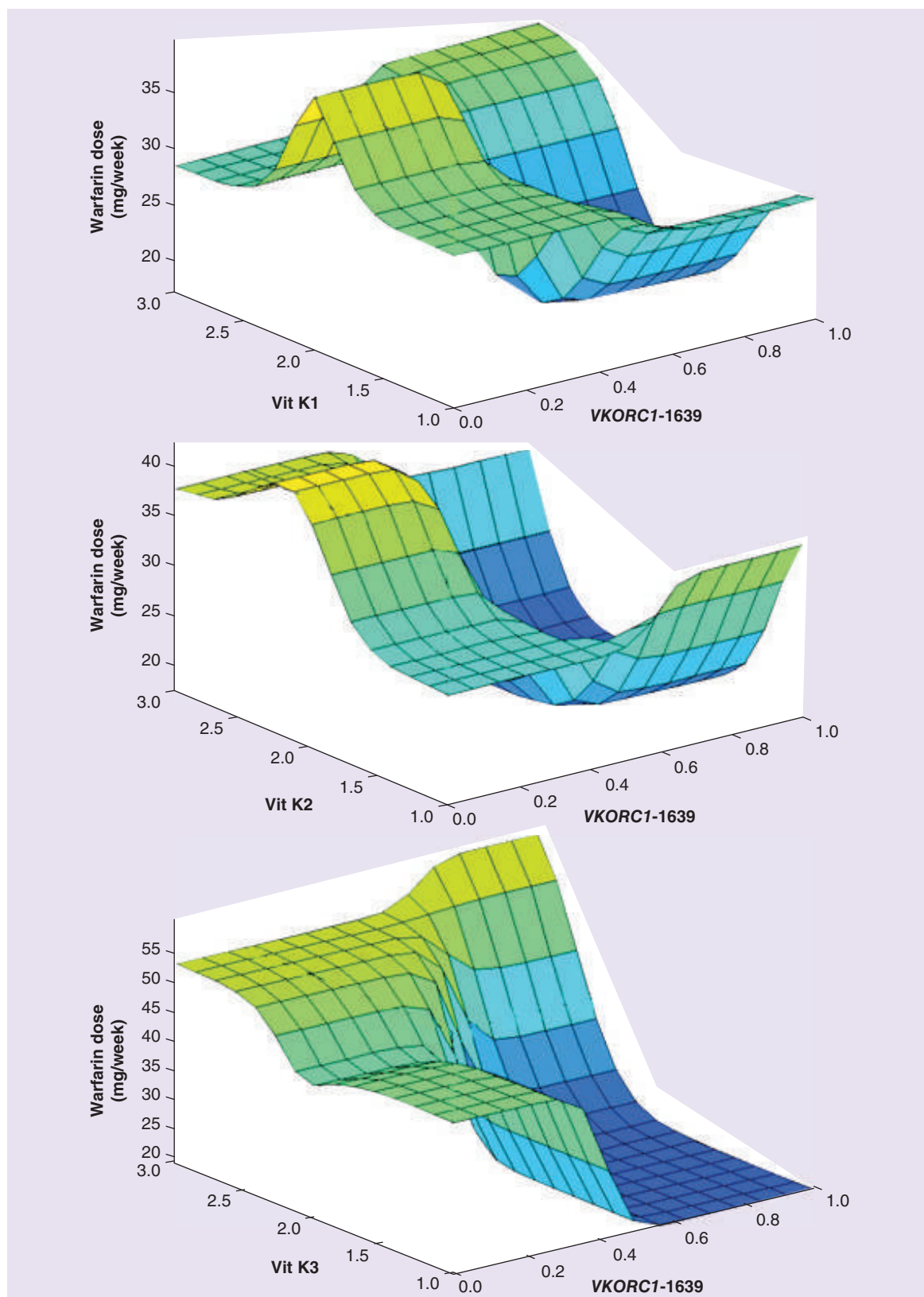


Figure 2. Association of *VKORC1*-1639G>A with warfarin in combination with varying levels of vitamin K1, K2 and K3. Irrespective of vitamin K1, K2 and K3, *VKORC1*-1639 G>A showed association with lesser warfarin dose requirement. However, with K3, positive association of warfarin dose was more prominent.

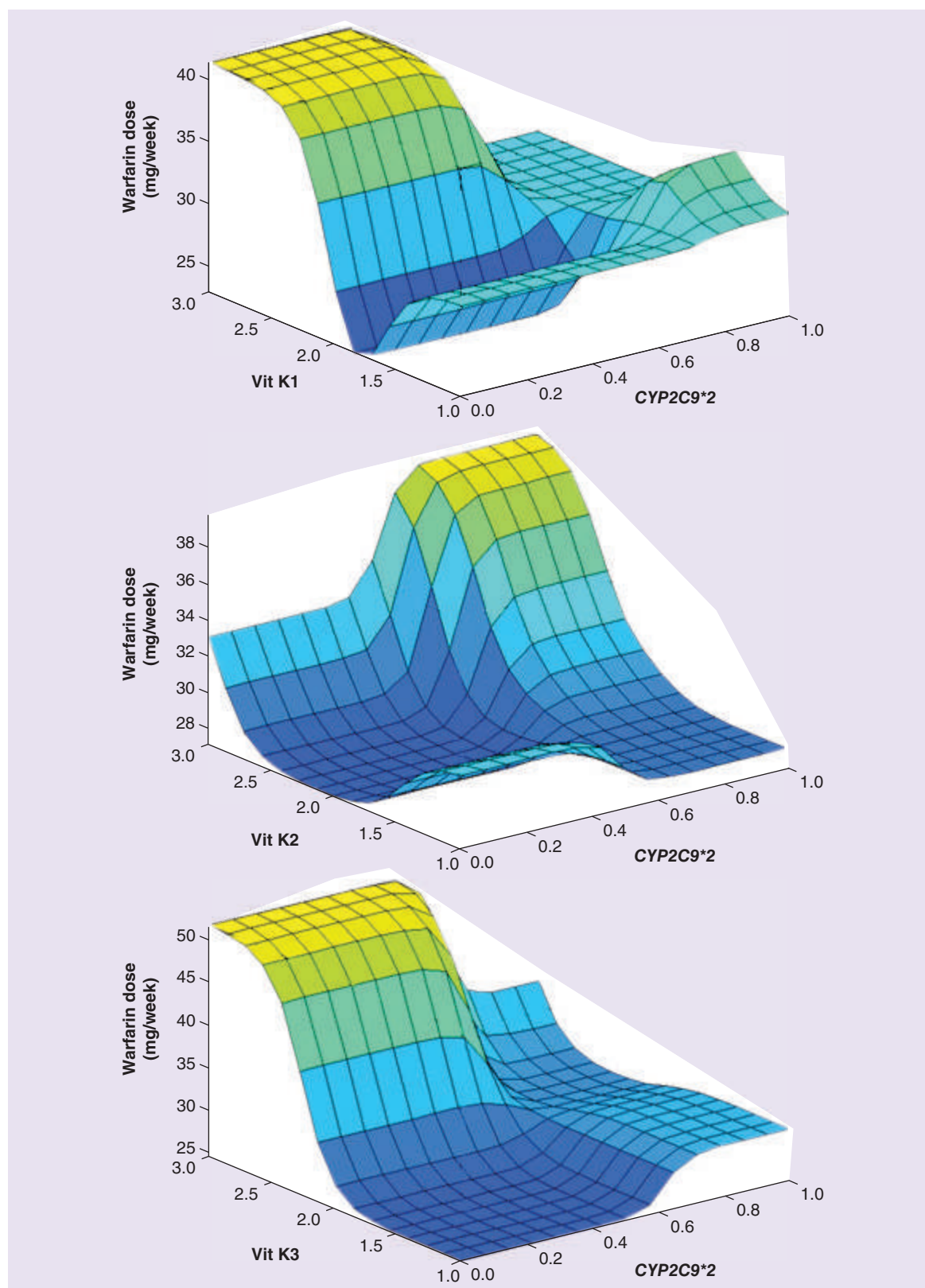


Figure 3. Association of warfarin dose with CYP2C9*2 and vitamin K1, K2, K3 levels. (A) CYP2C9*2 mediated warfarin sensitivity is more pronounced only in subjects with vitamin K1 in highest tertile. (B) CYP2C9*2 mediated warfarin sensitivity is observed in subjects with vitamin K2 levels in the first tertile. (C) CYP2C9*2 mediated warfarin sensitivity is observed in subjects with vitamin K3 levels in the first tertile.

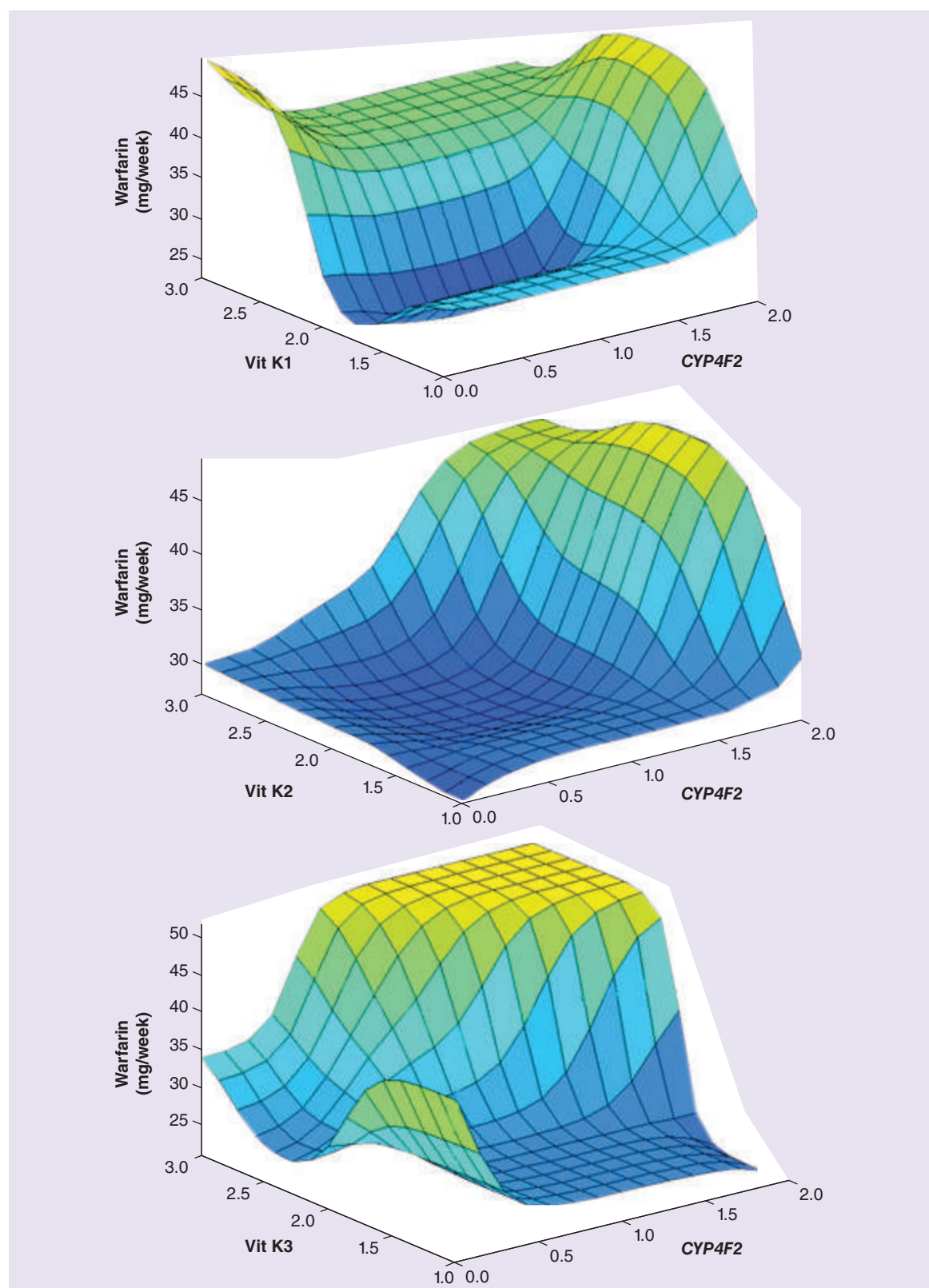


Figure 4. Association of warfarin dose with *CYP4F2* and vitamin K1, K2, K3 levels. (A) *CYP4F2* homozygous mutant exhibits higher requirement of warfarin dose when vitamin K1 levels are in second tertile. (B) *CYP4F2* homozygous mutant exhibits higher requirement of warfarin dose in all tertiles of vitamin K2 levels. (C) In *CYP4F2*, requirement of warfarin dose is more in second and third tertiles where there was higher vitamin K3 levels.

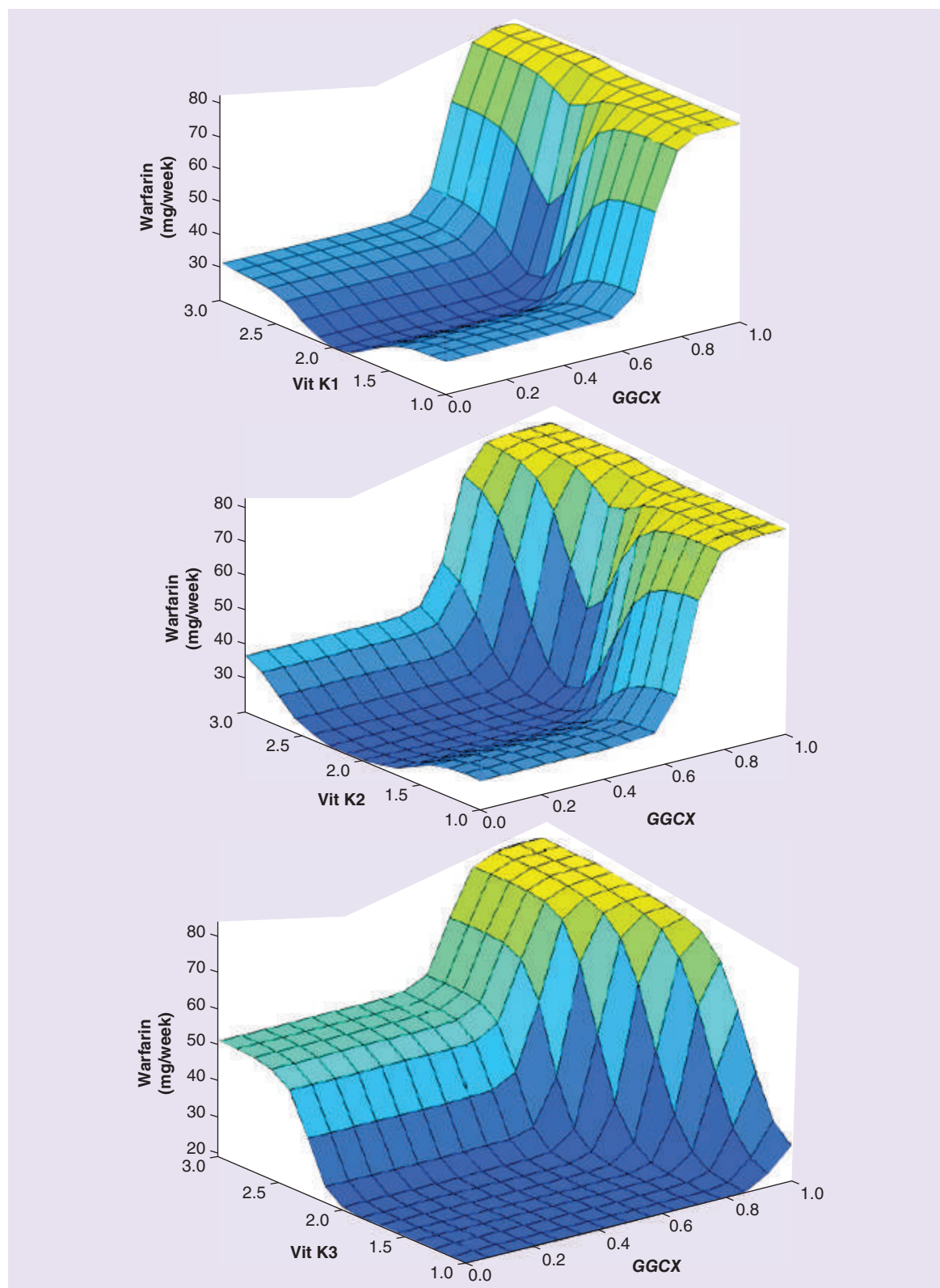


Figure 5. Association of warfarin dose with GGCX and vitamin K1, K2, K3 levels. (A) GGCX wild genotype exhibits low requirement of warfarin dose when vitamin K1 levels are in first and second tertile. Presence of variant allele increases warfarin dose proportional to vitamin K1. (B) GGCX wild type shows inverse association with requirement of warfarin dose. (C) GGCX mediated warfarin sensitivity is observed in third tertile where there was higher vitamin K3 levels.

shown to exhibit 44% increase in activity [24]. The warfarin sensitivity attributed to this polymorphism was consistent irrespective of vitamin K1, K2 and K3 levels. There was no significant association that was observed between the *VKORC1* polymorphism and frequency of excess of anticoagulation (INR >4.0) [25].

The *GGCX* 8016GG genotype was reported to exhibit higher mean activity of Protein C compared with AG and AA genotypes [26]. Since protein C acts as an anticoagulant, the requirement of warfarin as an anticoagulant might be lower in subjects with GG genotypes as observed in the current study irrespective of vitamin K1, K2 and K3.

The *CYP4F2* V433M Polymorphism was reported to reduce capacity to metabolize vitamin K1 and hence requiring higher warfarin dose to elicit effective anticoagulant response [27]. Our results corroborate with it.

In the presence of *CYP2C9* polymorphisms, S-warfarin is not hydroxylated to form 7-hydroxy warfarin and hence the biological availability of warfarin to inhibit *VKORC1* is more. *VKORC1* catalyzes the reduction of KO to vitamin K. However, the ability of *VKORC1* to further reduce vitamin K to KH2 was reported to be lesser *in vitro* [28]. *NQO1* catalyzes the reduction of vitamin K3 to menadiol hydroquinone (KH2 form of K3). Tie *et al.* have demonstrated the possibility of two enzymes that reduce vitamin K to KH2 with one being warfarin-sensitive and the other being warfarin resistant [29]. Future studies in this direction might help in identifying the mechanism of conversion of vitamin K1 and K2 to KH2 form. KH2 is the main cofactor required by *GGCX* to catalyze post-translational carboxylation of specific glutamic

acid residues to γ -carboxyglutamic acid (gla) to active coagulation factors.

The positive association between vitamin K intake and warfarin dose was reported in a recent study [30]. Supratherapeutic INR was observed in a patient whose vitamin K intake was low [31]. Usual vitamin K intake was shown to have no significant impact on the anticoagulant function of warfarin in a study with a smaller sample size [32].

In the two independent studies, the inverse correlation between plasma vitamin K/dietary vitamin K and INR was reported [33,34]. When there was a consistent intake of vitamin K it could reduce inpatient variability in anticoagulation response. The high intake of vitamin K was shown to reduce the therapeutic efficacy of warfarin, while low intake of vitamin K appears to be more sensitive [35]. The daily intake of vitamin K-rich food for one week was shown to raise the INR values above the therapeutic limit therefore requiring warfarin dose adjustment [36].

Till date, majority of the pharmacogenomic models addressed inter-individual variations in warfarin dose requirement based on the genetic constitution. The inclusion of nutrients, specifically vitamin K might help in addressing intraindividual variation as well. The current study addressed this issue and explained changes in warfarin dose requirement across the three tertiles of vitamin K1, K2 and K3 in all genotype combinations.

The limitations of current model were: Noninclusion of possible drug–drug interactions, and influence of hypo and hyper thyroidism on vitamin K dependent clotting factors. Future studies are warranted on dif-

Executive summary

Background

- Warfarin exerts anticoagulant effect by inhibiting the activation of vitamin K-dependent clotting factors.
- The existing pharmacogenomic algorithm explained interindividual variations in warfarin dose requirement based on demographic and genetic factors.
- There is a need to address the factors that contribute toward intraindividual variations in warfarin dose.

Materials & methods

- Patients receiving optimal dose of warfarin (n = 115) were analyzed for vitamin K1, K2 and K3 levels in plasma by HPLC, the genetic polymorphisms, in other words, *CYP2C9**2, *CYP2C9**3, *VKORC1*-1639G>A, *CYP4F2* V433M and *GGCX* G8016A were detected by PCR-RFLP.
- A neuro-fuzzy logic model was developed to elucidate bivariate gene–nutrient interactions contributing toward alterations in warfarin dose requirement.

Results & conclusion

- The mean absolute error of the developed neuro-fuzzy logic model was 0.000024 mg/week.
- *CYP2C9**2 and *CYP2C9**3 mediated warfarin sensitivity was observed when vitamin K levels are in high and low tertiles, respectively.
- *VKORC1*-1639G>A exhibited warfarin sensitivity irrespective of vitamin K1, K2 and K3 levels.
- In the presence of *CYP4F2* V433M polymorphism higher vitamin K1 levels were observed.
- *GGCX* 8016 GG genotype was shown to have a lesser warfarin requirement compared to GA and AA genotypes.

ferent ethnic groups and populations to replicate and confirm these associations.

Conclusion

The current study demonstrates a positive association between vitamin K levels and warfarin dose in the majority of genotype combinations. In the presence of *CYP4F2* rs2108622 polymorphism, significant elevation in vitamin K1 levels was observed, suggesting inability of *CYP4F2* in executing vitamin K1 oxidase function. GGCX GG genotype showed a lesser requirement of warfarin compared to AG and AA genotypes irrespective of vitamin K levels. *CYP2C9**2 and *CYP2C9**3 exhibited warfarin sensitivity when their vitamin K levels in the highest and lowest tertiles, respectively. *VKORC1*-1639G>A showed warfarin sensitivity across all the tertiles of vitamin K1 and K3. However, marked elevation in vitamin K2 levels reverts this association.

Future perspective

The neuro-fuzzy model developed in this study can be further strengthened by including information related

to thyroid status and ethnicity to generalize the performance of the model.

Financial and competing interests disclosure

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No writing assistance was utilized in the production of this manuscript.

Ethical conduct of research

The authors state that they have obtained institutional review board approval and followed the principals outlined in the declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

References

Papers of special note have been highlighted as: • of interest

- Rost S, Fregin A, Ivaskevicius V *et al.* Mutations in *VKORC1* cause warfarin resistance and multiple coagulation factor deficiency type 2. *Nature* 427, 537–541 (2004).
- Yamazaki H, Shimada T. Human liver cytochrome P450 enzymes involved in the 7-hydroxylation of *R*- and *S*-warfarin enantiomers. *Biochem. Pharm.* 54, 1195–1203 (1997).
- Rettie AE, Wienkers LC, Gonzalez FJ, Trager WF, Korzekwa KR. Impaired (*S*)-warfarin metabolism catalysed by the R144C allelic variant of *CYP2C9*. *Pharmacogenetics* 4, 39–42 (1994).
- Pavani A, Naushad SM, Stanley BA, Kamakshi RG, Abinaya K, Rao MA. Mechanistic insights into the effect of *CYP2C9**2 and *CYP2C9**3 variants on the 7-hydroxylation of warfarin. *Pharmacogenomics* 16, 393–400 (2015).
- The multiple linear regression model developed has good accuracy, higher specificity and moderate sensitivity. This model also helps in predicting the conversion of warfarin to 7-OH warfarin. The hydrogen bonding interactions were lost in *CYP2C9**2 and *CYP2C9**3 genes which contribute to reduced affinity and poor hydroxylation of warfarin.
- Scott SA, Edelmann L, Kornreich R, Desnick RJ. Warfarin pharmacogenetics: *CYP2C9* and *VKORC1* genotypes predict different sensitivity and resistance frequencies in the Ashkenazi and Sephardi Jewish populations. *Am. J. Hum. Genet.* 82, 495–500 (2008).
- International Warfarin Pharmacogenetics Consortium, Klein TE, Altman RB *et al.* Estimation of the warfarin dose with clinical and pharmacogenetic data. *N. Engl. J. Med.* 360, 753–764 (2009).
- A pharmacogenetic dose algorithm was developed that includes *VKORC1*, *CYP2C9* genotypes and clinical variables to predict stable therapeutic dose. This algorithm benefits the patients who require less than 21 mg/week or greater than 49 mg/week.
- Ferder NS, Eby CS, Deych E *et al.* Ability of *VKORC1* and *CYP2C9* to predict therapeutic warfarin dose during the initial weeks of therapy. *J. Thromb. Haemost.* 8, 95–100 (2010).
- Wadelius M, Chen LY, Eriksson N *et al.* Association of warfarin dose with genes involved in its action and metabolism. *Hum. Genet.* 121, 23–34 (2006).
- Pavani A, Naushad SM, Kumar RM, Srinath M, Malempati AR, Kutala VK. Artificial neural network-based pharmacogenomic algorithm for warfarin dose optimization. *Pharmacogenomics* 17, 121–131 (2016).
- The pharmacogenomic model developed explained 92.5% variability in warfarin dosing with 5.8% of underestimation rate and 0.4% overestimation rate. Performance of this model was excellent in both euthyroid and hypothyroid status.
- Shaw PB, Donovan JL, Tran MT, Lemon SC, Burgwinkle P, Gore J. Accuracy assessment of pharmacogenetically predictive warfarin dosing algorithms in patients of an academic medical center anticoagulation clinic. *J. Thromb.* 30(2), 220–225 (2010).
- Gage BF, Eby C, Johnson JA *et al.* Use of pharmacogenetic and clinical factors to predict the therapeutic dose of warfarin. *Clin. Pharmacol. Ther.* 84(3), 326–331 (2008).
- Zhu Y, Shennan M, Reynolds KK *et al.* Estimation of warfarin maintenance dose based on *VKORC1* (-1639 G>A) and *CYP2C9* genotypes. *Clin. Chem.* 53(7), 1199–1205 (2007).

- 13 Wei M, Ye F, Xie D *et al.* A new algorithm to predict warfarin dose from polymorphisms of *CYP4F2*, *CYP2C9* and *VKORC1* and clinical variables: derivation in Han Chinese patients with non valvular atrial fibrillation. *Thromb. Haemost.* 107(6), 1083–1091 (2012).
- 14 Pavani A, Naushad SM, Uma A, Kutala VK. Methodological issues in the development of a pharmacogenomic algorithm for warfarin dosing: comparison of two regression approaches. *Pharmacogenomics* 15(8), 1125–1132 (2014).
- 15 Zhou Q, Kwong J, Chen J *et al.* Use of artificial neural network to predict warfarin individualized dosage regime in Chinese patients receiving low-intensity anticoagulation after heart valve replacement. *Int. J. Cardiol.* 176(3), 1462–1464 (2014).
- 16 Saleh MI, Alzubiedi S. Dosage individualization of warfarin using artificial neural networks. *Mol. Diagn. Ther.* 18(3), 371–379 (2014).
- 17 Pirmohamed M, Burnside G, Eriksson N *et al.* A randomized trial of genotype-guided dosing of warfarin. *N. Engl. J. Med.* 369(24), 2294–2303 (2013).
- ***CYP2C9* and *VKORC1* polymorphisms together with BMI account for 50% of intraindividual variability in warfarin dosing. After initiation of warfarin therapy difference in INR was observed and there is a varied anticoagulation response. A new pharmacogenomic algorithm was developed which analyzes the effect of *CYP2C9* polymorphisms. This genotype-guided warfarin dosing is superior to standard dosing algorithms.**
- 18 Kimmel SE, French B, Kasner SE *et al.* A pharmacogenetic versus a clinical algorithm for warfarin dosing. *N. Engl. J. Med.* 369(24), 2283–2293 (2013).
- 19 Sato Y, Murata M, Chiba T, Umegaki K. A systematic review of the acceptable intake level of vitamin K among warfarin users. *J. Food Hyg. Soc. Jpn Food Hyg. Saf. Sci.* 56, 157–165 (2015).
- 20 Kim Y-E, Woo HI, On YK, Kim JS, Lee S-Y. High intra- and inter-individual variability of plasma vitamin K concentrations in patients with atrial fibrillation under warfarin therapy. *Eur. J. Clin. Nutr.* 69, 703–706 (2015).
- **Higher inter- and intraindividual variability in vitamin K was observed in warfarin treated patients. This shows that plasma vitamin K measurement might be helpful in predicting the dietary control and anticoagulation response.**
- 21 Leblanc C, Presse N, Lalonde G, Dumas S, Ferland G. Higher vitamin K intake is associated with better INR control and a decreased need for INR tests in long-term warfarin therapy. *Thromb. Res.* 134, 210–212 (2014).
- 22 Hirai K, Yamada Y, Hayashi H, Tanaka M, Izumiya K, Suzuki M. Plasma vitamin K concentrations depend on *CYP4F2* polymorphism and influence on anticoagulation in Japanese patients with warfarin therapy. *Thromb. Res.* 135, 861–866 (2015).
- **Patients with the *CYP4F2* TT genotype had significantly higher plasma VK1 and MK-4 concentrations than those with CC and CT genotypes. The multiple linear regression model developed explains 42% of variability in warfarin dosing. The *CYP4F2* polymorphism is poorly associated with inter-individual variability of warfarin dosage requirement.**
- 23 Crosier MD, Peter I, Booth SL, Bennett G, Dawson-Hughes B, Ordovas JM. Association of sequence variations in vitamin K epoxide reductase and γ -glutamyl carboxylase genes with biochemical measures of vitamin K status. *J. Nutr. Sci. Vitaminol.* 55, 112–119 (2009).
- 24 Yuan HY, Chen JJ, Lee MT *et al.* A novel functional *VKORC1* promoter polymorphism is associated with inter-individual and inter-ethnic differences in warfarin sensitivity. *Hum Mol Genet.* 14(13), 1745–1751 (2005).
- 25 Rasmussen MA, Skov J, Bladbjerg E-M, Sidelmann JJ, Vamosi M, Jespersen J. Multivariate analysis of the relation between diet and warfarin dose. *Eur. J. Clin. Pharmacol.* 68, 321–328 (2011).
- 26 Kimura R, Miyashita K, Kokubo Y *et al.* Genotypes of vitamin K epoxide reductase, gamma-glutamyl carboxylase, and cytochrome P450 2C9 as determinants of daily warfarin dose in Japanese patients. *Thromb. Res.* 120(2), 181–186 (2007).
- 27 McDonald MG, Rieder MJ, Nakano M, Hsia CK, Rettie AE. *CYP4F2* is a vitamin K1 oxidase: an explanation for altered warfarin dose in carriers of the V433M variant. *Mol. Pharmacol.* 75(6), 1337–1346 (2009).
- 28 Chu PH, Huang TY, Williams J, Stafford DW. Purified vitamin K epoxide reductase alone is sufficient for conversion of vitamin K epoxide to vitamin K and vitamin K to vitamin KH2. *Proc. Natl Acad. Sci. USA.* 103(51), 19308–19313 (2006).
- 29 Tie JK, Jin DY, Straight DL, Stafford DW. Functional study of the vitamin K cycle in mammalian cells. *Blood* 117(10), 2967–2974 (2011).
- 30 Reaves AB, Clarke CJ, Tillman EM. Supratherapeutic international normalized ratio due to reduced vitamin K intake secondary to prolonged vomiting in a patient on warfarin. *Ann. Pharmacother.* 47(6), e28 (2013).
- 31 Chertovskikh YV, Malova EU, Maksimova NR, Popova NV, Sychev DA. *VKORC1* polymorphisms and warfarin maintenance dose in population of Sakha. *Int. J. Risk Saf. Med.* 27(Suppl. 1), S17–S18 (2015).
- 32 Park JN, Lee JS, Noh MY, Sung M-K. Association between usual vitamin K intake and anticoagulation in patients under warfarin therapy. *Clin. Nutr. Res.* 4, 235 (2015).
- 33 Khan T, Wynne H, Wood P *et al.* Dietary vitamin K influences intra-individual variability in anticoagulant response to warfarin. *Br. J. Haematol.* 124, 348–354 (2004).
- 34 Couris R, Tataronis G, McCloskey W *et al.* Dietary vitamin K variability affects international normalized ratio (INR) coagulation indices. *Int. J. Vit. Nutr. Res.* 76, 65–74 (2006).
- 35 Johnson MA. Influence of vitamin K on anticoagulant therapy depends on vitamin K status and the source and chemical forms of vitamin K. *Nutr. Rev.* 63, 91–97 (2005).
- 36 Karlson B, Leijd B, Hellström K. On the influence of vitamin K-rich vegetables and wine on the effectiveness of warfarin treatment. *Acta Medica Scandinavica.* 220, 347–350 (2009).